

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-08

BOTANY

ARSHID BHAT SIR



Topics To Covered

1 C₄ Pathway

2 ...

Key Points Regarding Photorespiration

- Photorespiration is a wasteful process

 - No Sugare is formed.
 - No ATP & $\text{NADPH} + \text{H}^+$ formed.
 - ATP is Used.
- Photorespiration occurs only in C_3 plants but not in C_4 plants. Thus C_4 plants are photosynthetically more efficient as compared to C_3 plants.

C₄ Pathway

- It is also called as Hatch and Slack pathway.
- C₄ plants are adapted to dry tropical regions.
- About 1% of total plants on Earth are C₄ plants.

• Examples:

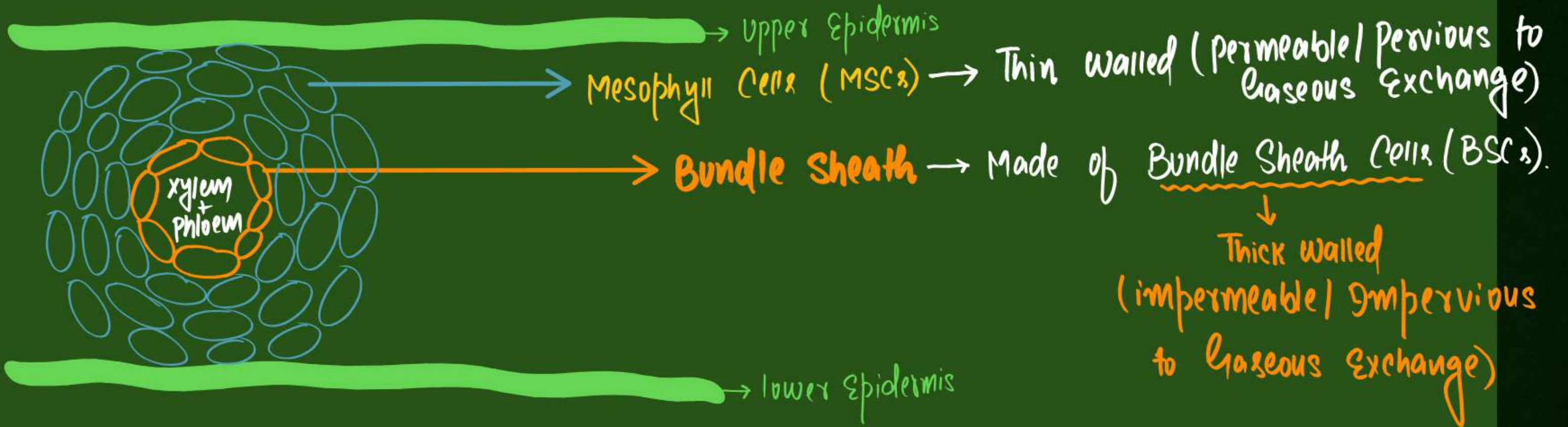
Maize, Sorghum, Sugarcane → Monots.

Amaranthus, Chenopodium, Portulaca → Dicots.

NOTE C₄ plants possess special type of leaf anatomy called as Kranz Anatomy

Kranz Anatomy

- Kranz means wreath / ring like.

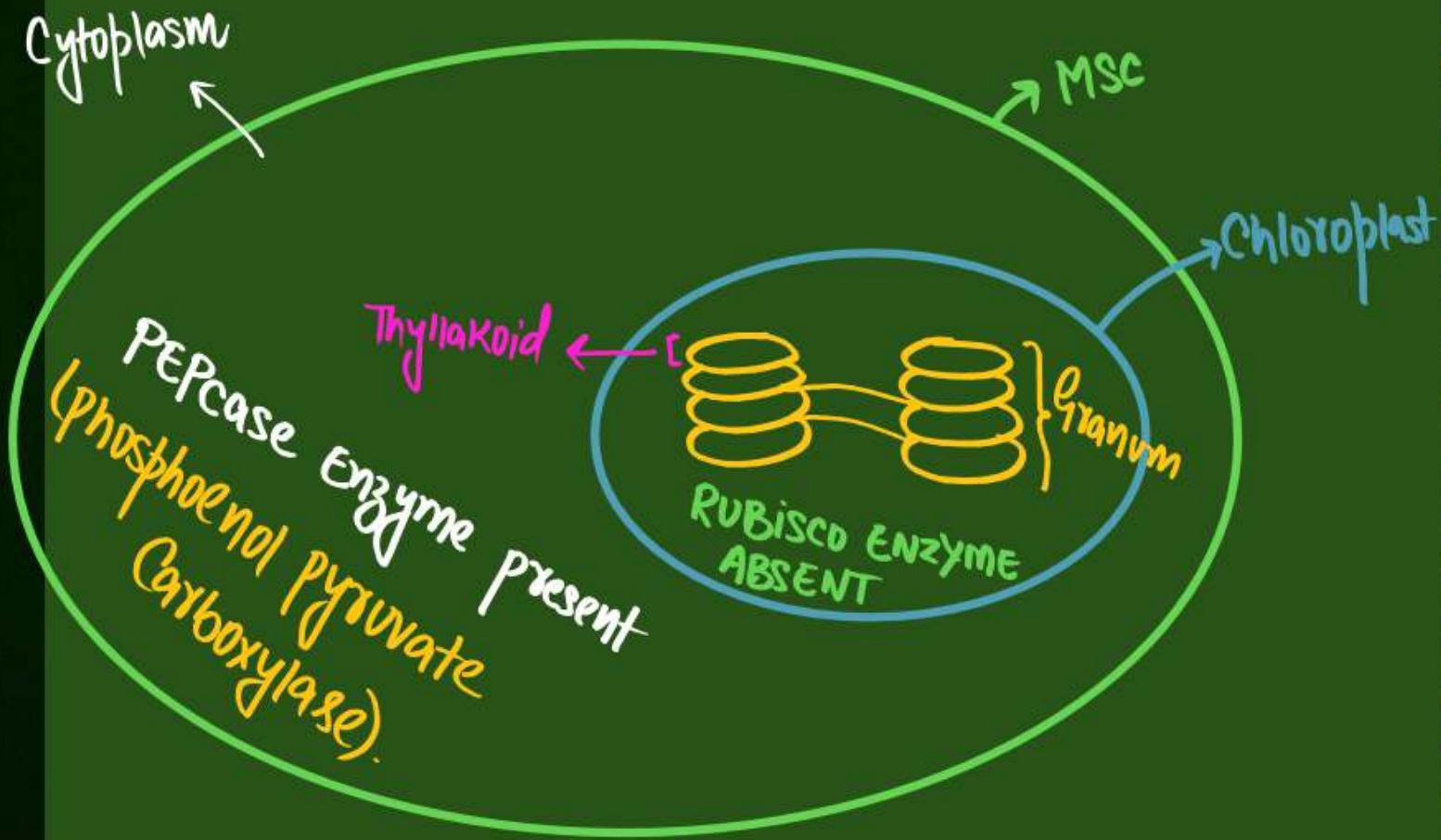


- ↳ In Kranz anatomy mesophyll cells are clustered around Bundle sheath cells in ring/wreath manner.
- ↳ Bundle sheath cells are several layered thick.
- ↳ Mesophyll cells and Bundle sheath cells are interconnected by means of plasmodesmata connections.
- ↳ Here Mesophyll is not differentiated into palisade and Spongy parenchyma.

NOTE • In C₄ plants, Chloroplast is dimorphic.

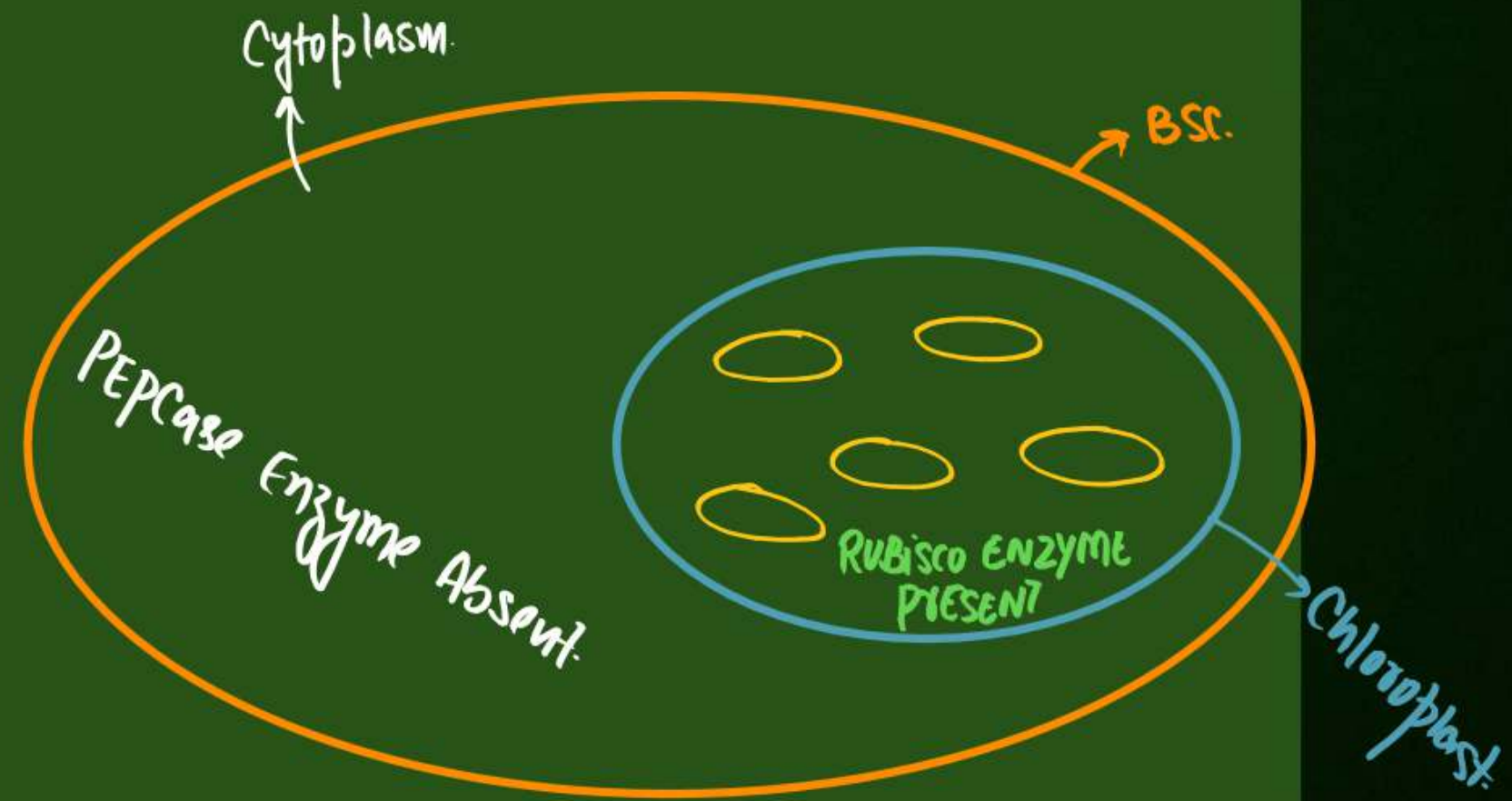
Granal Chloroplast → Grana present

• In Mesophyll cells (MSCs)



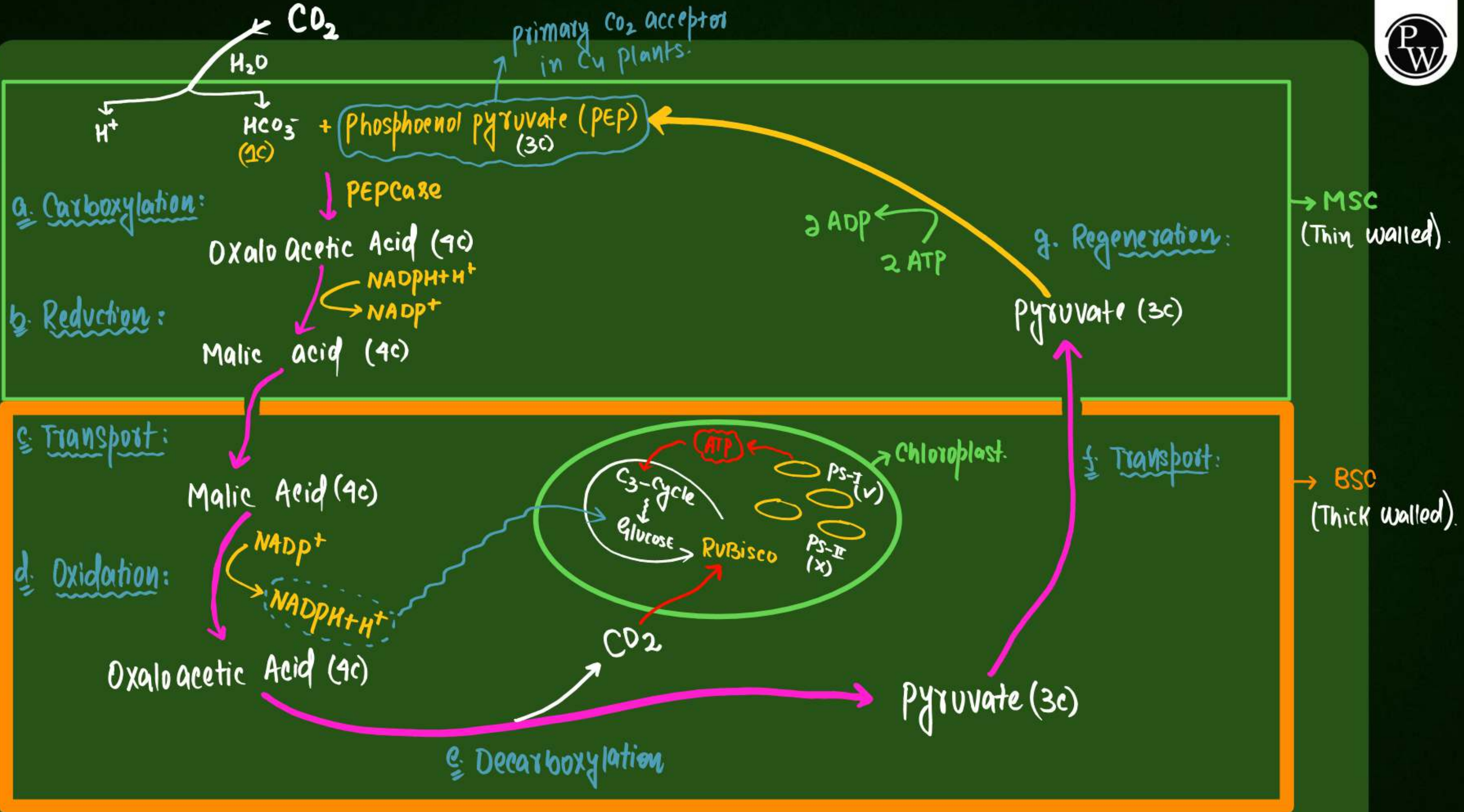
Agranal Chloroplast → Grana Absent.

• In Bundle Sheath cells (BSCs)



NOTE

• C₄ Cycle involves both Cells i.e, Mesophyll cells
and
Bundle Sheath cells.



Important Points

①: C_4 -pathway.

INPUT	OUTPUT
$6 \times CO_2$ $30 \times ATP \begin{cases} \nearrow \text{Regeneration} = 12 ATP \\ \searrow C_3\text{-cycle} = 18 ATP \end{cases}$ $12 \times NADPH + H^+$	$1 \times C_6H_{12}O_6$ $30 \times ADP$ $12 \times NADP^+$

② C_4 plants tolerate H_2O Stress and high temp conditions bacuse.

a. leaf posses thick Cuticle.

b. presense of Sunken Stomata.

③. Photorespiration does not occur in C_4 plants and thus produce greater amount of biomass as compared to C_3 -plants.

④. In C_4 -plants, CO_2 for C_3 -cycle is produced intracellularly (within the cell) by decarboxylation of Oxaloacetic acid (OAA) to pyruvate.

⑤ C_3 -cycle $\left\{ \begin{array}{l} \text{occurs in Msc (in } C_3 \text{ plants)} \\ \text{occurs in Bsc (in } C_4 \text{ plants)} \end{array} \right.$

- ⑥ Primary CO_2 acceptor in C_4 plants is phosphoenol pyruvate (3C).
- ⑦ First product formed in C_4 -cycle is Oxalo acetic acid (OAA) which is 4-C compound i.e., why the name C_4 -cycle.
- ⑧. phosphoenol pyruvate Carboxylase
(PEPcase) → performs Carboxylation even in low concentrations of CO_2 .

Factors Affecting Rate Of Photosynthesis

↓
NEXT lecture

QUESTIONS

Given below are two statements:

[MR] (2023-Manipur)

Statement-I: RuBisCO is the most abundant enzyme in the world.

Statement-II: Photorespiration does not occur in C_4 plants. In the light of the above statements, choose the most appropriate answer from the options given below:

- A** Statement-I is correct but Statement-II is incorrect
- B** Statement-I is incorrect but Statement-II is correct
- C** Both Statement-I and Statement-II are correct
- D** Both Statement-I and Statement-II are incorrect

M/W

QUESTIONS

What is the role of large bundle sheath cells found around the vascular bundles in C_4 plants? (2022)

- A** To protect the vascular tissue from high light intensity
- B** To provide the site for photorespiratory pathway
- C** To increase the number of chloroplast for the operation of Calvin cycle
- D** To enable the plant to tolerate high temperature

QUESTIONS

H/w

Given below are two statements:

(2022)

Statement-I: The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells.

Statement-II: Mesophyll cells of C_4 plants lack RuBisCO enzyme

In the light of the above statements, choose the correct answer from the options given below.

- A** Statement-I is incorrect but Statement-II is correct
- B** Both Statement-I and Statement-II are correct
- C** Both statement-I and statement-II are incorrect
- D** Statement-I is correct but Statement-II is incorrect

QUESTIONS

M/W



In C_4 plants, Calvin cycle operates in:

(1989)

- A** Stroma of bundle sheath chloroplasts
- B** Grana of bundle sheath chloroplasts
- C** Grana of mesophyll chloroplasts
- D** Stroma of mesophyll chloroplasts

HOMEWORK

3 Questions



Thank
You